Teacher only: do not distribute to students

Distribution and research administration

1. Distribute A first. Preserve the original hypothesis, reasons, variables, assumptions, control, drawing and repeat plan. Keep a readable student copy before distributing B.
2. B uses three independent plates under shared teaching conditions of 30°C and 24 hours. Students interpret the images and raw data, record observations, calculate means and complete the six analysis items.
3. Collect or fix the original answers in A and B before distributing C. Reflection uses the student’ s own experimental results. C prints no reference conclusion or correctness marks.
4. D and the complete pack are for teachers only. Do not distribute the entire pack to students at once. Keep language, available time and prompts consistent（一致的） across groups.
5. This feature does not read or create student records or change answers, drawings, demonstrations, timing, events, cloud records or Excel. Researchers separately preserve and score handwritten paper data.
6. The supplied proposal, sections 4.1 and 4.3.2, specifies four weekly modules before formal teaching, with identical scenarios, concepts and prompts. Paper replaces interactive simulation with static diagrams and pre-rendered data tables. This booklet corresponds to VL4 antibiotics（抗生素）.
7. Paper uses fixed seeds; individual VL inquiries use independent seeds. The parameters and result model are the same, but individual raw values may differ. The researcher must confirm that this variation fits the study design.
8. Paper cannot measure virtual tool operation. Review cross-group SPS coverage, the 23-mark weighting, inter-rater agreement and separate formal SPS/concept pre/post tests. Trial-print B in greyscale and check image and text readability before use.
9. The proposal specifies 20 students per group, combining X1/X2: 11/9 in VL and 12/8 in paper. The separate 10-item SPSAI and 4-item CKT are administered in week 0 and after week 4, using the same pre/post tests. This worksheet does not replace those instruments or establish their validity.

These draft criteria have not been validated or calibrated across VLs. Speed, clicks, completion rate or a prediction matching the outcome are not measures of ability.

Images and raw diameters are generated by the current VL4 result model. No colony（菌落） counts, drug concentrations or patient doses have been added. Raw data are not mean or conclusion answers.

VL1 reference: v3 · No digestive-system questions are imported.

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Reference answers

Independent variable	Sample in the disc
Dependent variable	Visible clear zone（清晰區） and its total diameter
Controlled variables	Disc size ／ Composition of the agar plate（瓊脂板） ／ Type of bacteria used ／ Initial bacterial distribution ／ Preset sample preparation
What are the assumptions of this investigation?	Bacteria are initially distributed evenly on each agar plate（瓊脂板）. ／ Equipment does not introduce contamination. ／ All plates use the same simulated observation conditions.
1. What does the appearance of a clear zone（清晰區） indicate?	B. Under these test conditions, there is no visible bacterial growth in the zone, indicating inhibition（抑制）.
2. What does the diameter of a clear zone（清晰區） represent?	C. The extent of the area with no visible bacterial growth under these conditions; diffusion and other factors also affect the diameter.
4. Why do experiments need to be repeated?	C. To reduce random error and assess the reproducibility（可重複性） of the results and the reliability（可靠性） of the data.
5. If repeated results differ greatly, what should you do, and why?	D. First check spreading, disc positions and plate conditions, then repeat as needed to identify causes of variation and improve the comparison.
6. When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	C. Provide an antibiotic（抗生素）-free comparison to help assess whether clear-zone（清晰區） formation is associated with the antibiotic（抗生素） sample.

Raw diameter data (mm)

Plate	X	Y	Z	C (control)
1	18.7	0.0	26.1	0.0
2	18.7	0.0	26.3	0.0
3	18.7	0.0	26.5	0.0
Mean total diameter calculation	18.7	0.0	26.3	0.0

Visible zone outside the disc? Yes：X、Z；No：Y、C (control)。

Draw a conclusion from your results: X、Z。

Comparing the mean total clear-zone（清晰區） diameters in this test, their inhibitory（抑制） effects rank as follows: Z ＞ X ＞ Not applicable ＞ Not applicable。

Score ranking against the student’ s own means; equal means allow either order.

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Written responses: reference guidance

Your initial observations

A clear zone（清晰區） with no visible bacterial growth is present near Penicillium（青黴菌）(a fungus). This is an observation; its cause needs testing.

Which samples do you predict will have an area without visible bacterial growth around the disc?

A reasonable testable prediction may differ from the results. A reasoned prediction does not have to match the outcome.

Which sample do you predict will have the largest clear zone（清晰區） around its disc?

Give a reasoned prediction based on diffusion and bacterial growth. Choose not applicable if you predict no clear zones（清晰區）. The prediction need not match the result.

Your reason (Hint: antibiotics（抗生素） can diffuse through the agar plate（瓊脂板）.)

Explain a reason related to bacterial growth and observable evidence.

If you want to find out whether antibiotics（抗生素） affect bacterial growth, how would you design a control group?

Use a matching carrier-only disc without antibiotic（抗生素） to check carrier/disc effects, with other conditions the same. Call it a water control only if every carrier is water.

Setup labels / written design

Label each sample and control and allow enough space between discs and from the edge to avoid overlapping zones. A drawing or written setup is acceptable.

There are 4 test samples: X, Y, Z and the control. How many tests do you think each sample needs in total? (Include the first test.)

Include the first test. Three is not a required correct answer; more repeats do not guarantee reliability（可靠性）.

Why would you plan it this way?

Consider consistency（一致性）, variation, fair comparisons and practical feasibility.

Choose an antibiotic（抗生素） sample from your experimental results that can inhibit（抑制） bacterial growth, and cite readings to support your judgement. Suppose this bacterial population initially contains a small number of bacteria resistant（具抗藥性的） to that antibiotic（抗生素）. How might its effectiveness change with continued use? Use the learning points to explain how bacterial survival, reproduction（繁殖） and inheritance（遺傳） of resistance traits（抗藥性特徵） cause this change.

Use readings for your chosen sample to support growth inhibition（抑制）. Explain how pre-existing resistance（抗藥性） affects survival and reproduction（繁殖）, how resistance traits（抗藥性特徵） are inherited（遺傳）, and how the resistant（具抗藥性的） proportion increases as antibiotic（抗生素） effectiveness declines. This is a hypothetical extension, not evidence that resistance（抗藥性） arose in this experiment.

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Scoring criteria (current VL4 draft)

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Item	Marks	Method	Full-credit criterion	Partial credit	Zero credit / limits
Initial observation	2	Teacher	Distinguish visible colonies（菌落） and the clear area near the fungus; describe observations without assuming a cause.	1: relevant description, but location or observation/inference distinction is incomplete.	No relevant observation, or only an unsupported causal claim.
Visible zones and diameters	2	Shared check	For each of 12 readings, use the correct Yes/No classification and a total diameter within ±1 mm; no visible zone requires exactly 0. Score matched readings × 2/12, rounded to two decimal places.	Proportional credit for matched readings.	No matching evidence; a 6 mm disc alone is not a clear zone（清晰區）.
Independent, dependent and controlled variables	4	Shared check	Select only sample for IV, zone for DV; controls are disc size, agar composition, strain, initial distribution and preset preparation.	IV: 1; DV: 1. Each correct control earns 2/5, rounded to two decimal places; an unrelated control choice gives 0 for the control item.	IV/DV must match the exact set.
Testable hypothesis and reason	1	Teacher	A testable prediction linked to bacterial growth and observable evidence.	0.5: a testable prediction with a weak or unclear reason.	No testable prediction or relevant reason. An unsupported prediction is not automatically wrong.
Original repeat-plan reason	1	Teacher	Explain repeats through consistency（一致性）, variation, fair comparison and feasibility.	0.5: mentions reliability（可靠性） but lacks independent comparison or a rationale for the number.	Only a number or no relevant reason; choosing three is not a required answer.
Control design and reason	1	Teacher	Use a matching carrier-only disc without antibiotic（抗生素）; hold other conditions constant and explain the comparison.	0.5: suitable C, but the reason or shared conditions are incomplete.	No useful comparison or confounded treatment.

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Item	Marks	Method	Full-credit criterion	Partial credit	Zero credit / limits
Assumptions	1	Shared check	Select only even initial distribution, no added contamination and shared observation conditions.	No partial credit.	Any missing or additional choice, including assuming all bacteria died.
Plate-design quality	2	Teacher	Label X, Y, Z and C clearly; allow space between discs and from the edge. A drawing or complete written design is acceptable.	1: basically feasible but missing one important arrangement.	No assessable design; artistic quality is not a criterion.
Means	1	Shared check	Use the student’s own three latest readings; ±0.1 mm tolerance; 0.25 per sample.	Proportional credit across four samples.	Do not penalise the same measurement error again through arithmetic.
Reflection: evidence and resistance（抗藥性） selection	2	Teacher	Cite sample readings to support inhibition（抑制）; link prior resistance（抗藥性） differences, survival and reproduction（繁殖）, inherited（遺傳） traits and changing population proportions to loss of effectiveness.	1: a reasonable experimental judgment or selection explanation, but missing readings or causal links.	No relevant evidence/concept, or bacteria purposefully developing resistance（抗藥性） to survive.
Analysis 1: What does the appearance of a clear zone（清晰區） indicate?	1	Shared check	Correct option: B.	No partial credit.	Incorrect or missing response: 0.
Analysis 2: What does the diameter of a clear zone（清晰區） represent?	1	Shared check	Correct option: C.	No partial credit.	Incorrect or missing response: 0.

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Item	Marks	Method	Full-credit criterion	Partial credit	Zero credit / limits
Analysis 4: Why do experiments need to be repeated?	1	Shared check	Correct option: C.	No partial credit.	Incorrect or missing response: 0.
Analysis 5: If repeated results differ greatly, what should you do, and why?	1	Shared check	Correct option: D.	No partial credit.	Incorrect or missing response: 0.
Analysis 6: When comparing the clear zones（清晰區） around C (control) and X, Y and Z, what is the main role of C?	1	Shared check	Correct option: C.	No partial credit.	Incorrect or missing response: 0.
Selected conclusion	1	Shared check	X and Z: 0.5. Rank effective samples by the student’ s own means: 0.5. Unused ranks are Not applicable.	0.5 if either component is correct.	Either order is accepted for equal means; do not penalise measurement error again.
Current total / pending marks	23	Formula	Total 23. All six manual fields must be scored and reflection submitted before a total is reported.	Blank manual marks mean pending; 0 means marked zero.	Do not compare directly with the legacy 32-mark rubric. Draft calibration is required.